

Large Language Models: Enterprise Applications

Executive Summary

Large Language Models (LLMs) represent a transformative shift in how organizations process and extract value from unstructured data.

KEY FINDINGS

1. Retrieval Augmented Generation (RAG)

RAG systems combine vector search with LLM generation.

They allow models to answer questions from private documents.

Accuracy improves by 40% compared to standalone LLMs.

2. Document Intelligence

Modern document AI can process PDF, images, and audio files.

OCR accuracy has reached 98.5% for printed text.

Multilingual models support 50+ languages simultaneously.

3. Enterprise Adoption

72% of Fortune 500 companies are piloting document AI.

Average ROI is 340% over 3 years.

Primary use cases: legal, finance, HR, and compliance.

4. Vector Databases

ChromaDB, Pinecone, and Qdrant lead the market.

Embedding dimensions range from 384 to 1536.

Cosine similarity is the standard retrieval metric.

RECOMMENDATIONS

Organizations should prioritize PII detection before ingestion.

Multi-tenant architectures are essential for enterprise deployment.

MLOps pipelines enable continuous model quality monitoring.

CONCLUSION

Document intelligence platforms like OmniDoc represent the next generation of enterprise knowledge management systems.